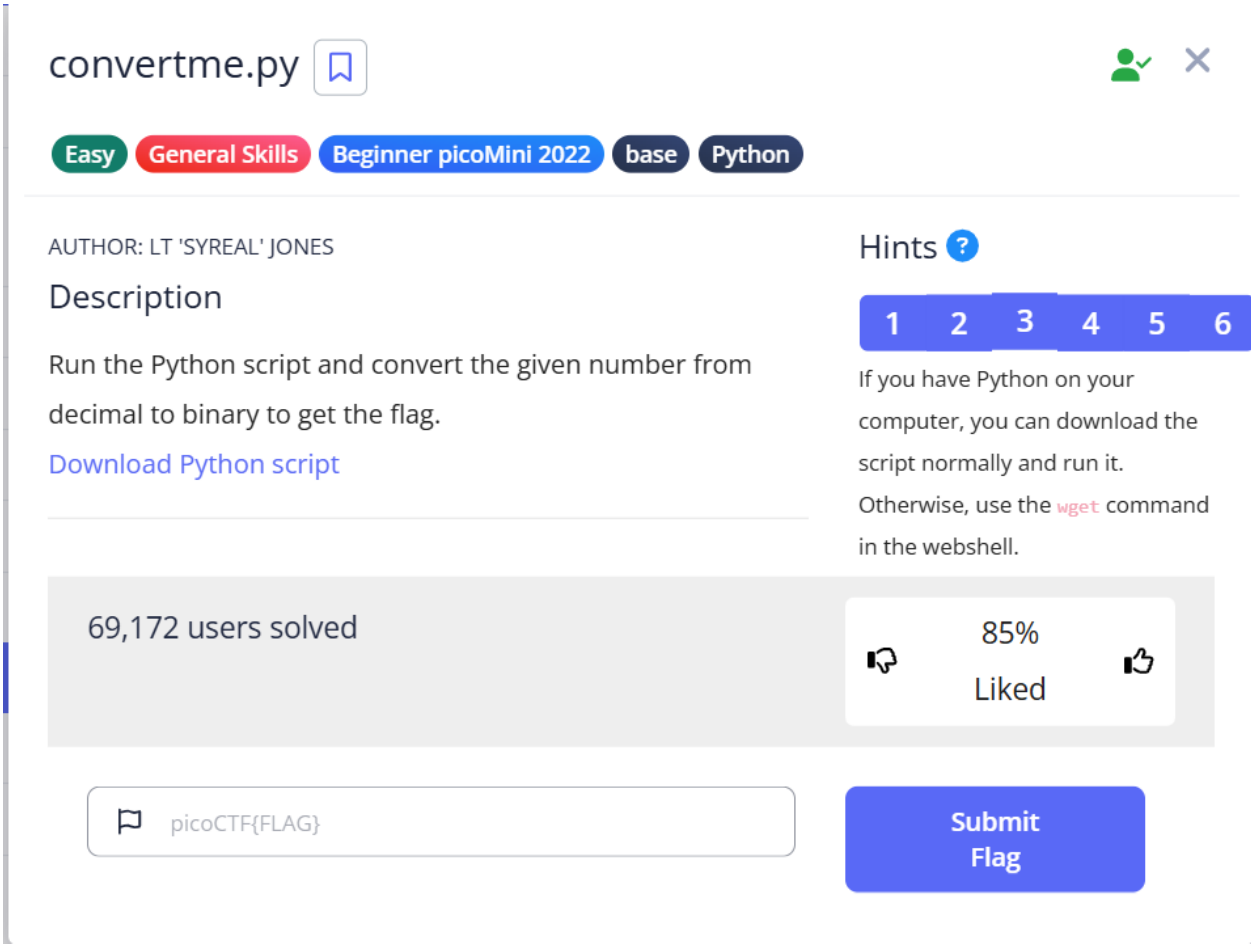
Saint Columban College
College of Computing Studies
Pagadian City

Convertme.py



Author: The Analyst: Hyposelenia

Challenge: Convertme.py

Category: General Skills

Date: 01/15/26

I. Objective

The objective of this challenge was to run a provided Python script, answer a number base–conversion question correctly, and retrieve the flag displayed by the program.

II. Background

In computer science and cybersecurity, numbers are often represented in different bases, such as decimal (base 10) and binary (base 2). Understanding how to convert between these number systems is an essential fundamental skill, especially when dealing with low-level data, encodings, and scripts that expect numeric input.

This challenge reinforces basic **decimal-to-binary conversion** while also practicing running Python programs in a Linux environment.

III. Tool Used

- **Oracle VirtualBox (Kali Linux) –** Used as the Linux environment
- **Linux Terminal –** Used to download and execute the Python script
- **wget –** Used to download the challenge file
- **Python 3 interpreter –** Used to run the script

IV. Methodology

1. Opened Oracle VirtualBox running Kali Linux.
2. Copied the download link of the Python file provided by the challenge.
3. Used the command wget <link address> to download the script.
4. A Executed the script using the command:

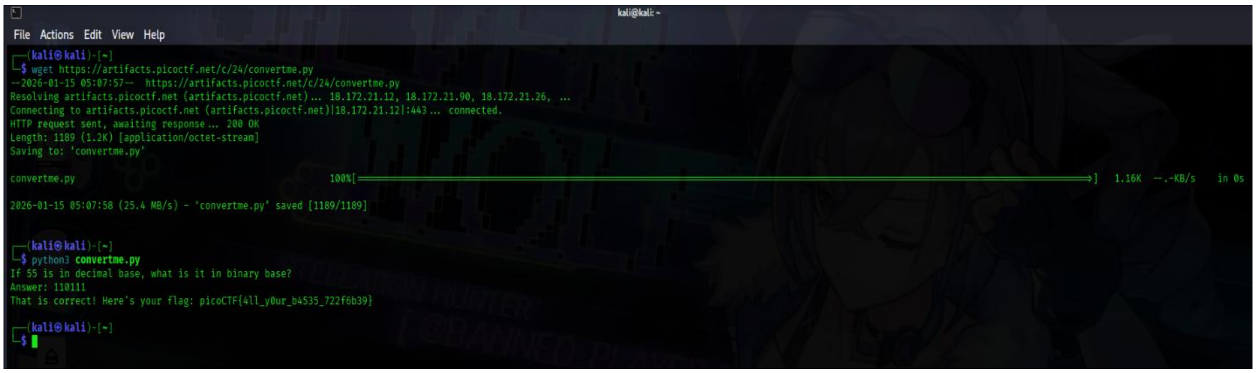
   python3 convertme.py

5. After running the program, a question appeared asking for the binary representation of the decimal number **55**.

6. Converted 55 from decimal to binary (110111) and entered the correct answer.
7. The program then printed the flag.

V. Result

picoCTF{4ll_y0ur_b4535_722f6b39}



VI. Explanation

The program prompts the user with a simple base-conversion question to ensure understanding of number systems. Decimal **55** converts to binary as follows:

- **55 ÷ 2 → remainder 1**
- **27 ÷ 2 → remainder 1**
- **13 ÷ 2 → remainder 1**
- **6 ÷ 2 → remainder 0**
- **3 ÷ 2 → remainder 1**
- **1 ÷ 2 → remainder 1**

Reading the remainders from bottom to top gives **110111**.

P.S.

It is possible to download the file and inspect the Python code directly. In some challenges, the flag may already exist in the script in an encoded or encrypted form. However, this would require reverse-engineering or decoding the logic to extract it manually.

In this challenge, running the program and correctly answering the prompt is the intended and simplest solution. I chose to use the Linux terminal to practice command-line usage and Python execution, which are essential skills in CTFs and real-world cybersecurity work. -xoxo

VII. Conclusion

This challenge reinforces basic number-system conversion while introducing or strengthening familiarity with running Python scripts in a Linux environment. It demonstrates that even simple foundational knowledge—such as binary conversion—plays an important role in cybersecurity challenges. Practicing both manual reasoning and command-line execution builds strong problem-solving habits for future, more complex tasks.

— **The Analyst: Hyposelenia**